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**TagSakay: A Low-Cost RFID Tapping System
for Monitoring Tricycle Queue Rotations in Davao City**

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I - INTRODUCTION

1.0 Project Context

Globally, rapid urbanization is exerting immense pressure on cities, driving the need for more efficient and organized public transportation systems, often leading to the adoption of innovative solutions to address complex urban mobility challenges [1]. The push for smart cities and digital governance emphasizes solutions that are not only efficient but also transparent and equitable for all stakeholders, including transport operators and commuters, by leveraging information and communication technologies to enhance public value and urban development [2].

In the Philippines, tricycle transport is a vital mode of public transportation in numerous local barangays, including Barangay Communal in Davao City. This sector plays a crucial role in last-mile connectivity and local commerce, supporting the daily lives of countless Filipinos. Recognizing its significance, there are national efforts to formalize and protect the interests of tricycle drivers and operators, aiming to ensure their dignified existence and economic advancement through proper regulation and support [3]. Despite its essential role, the current system for managing tricycle queue rotations in many areas, including Barangay Communal, often relies on traditional, manual processes such as chalkboard listings, verbal coordination, or simple memory. These outdated methods are highly prone to human error, miscommunication, and disputes among drivers, leading to

inefficiencies and fairness issues in the dispatch system.

As urbanization increases and digital governance continues to gain traction, grassroots-level transport systems require innovations that are affordable, easy to implement, and resilient to infrastructure limitations. The absence of an automated, verifiable queuing mechanism has highlighted the need for a system that can ensure order, transparency, and accountability, especially during peak hours, holidays, or in the presence of multiple routes.

The current queueing system in Barangay Communal specifically depends on informal methods such as chalkboard listings and verbal arrangements among drivers. These manual practices are highly prone to errors, subjective decision-making, and disputes, particularly during rush hours, holidays, or when multiple units arrive simultaneously. Inconsistencies in maintaining the order of arrival often result in arguments among drivers, while barangay personnel lack the tools to validate or audit queue entries effectively. Furthermore, there is no centralized or digital record of queue rotations, making it difficult for local authorities to manage terminal operations, review disputes, or gather insights for planning. This lack of structure undermines fairness, accountability, and operational efficiency.

The TagSakay project addresses these issues by proposing a digital tap-in system using RFID cards. This will enable real-time monitoring of tricycle queue rotations, minimize human error, and provide a visual display and backend dashboard

to facilitate transparent queueing practices.

While alternatives like GPS-based tracking or smartphone applications were considered, they introduce higher initial costs, require consistent driver smartphone ownership and data plans, and face challenges with signal reliability in shaded terminal areas. RFID technology, however, offers a robust, low-cost, and user-friendly solution. Its passive nature ensures minimal power consumption, its physical tap provides clear, verifiable timestamps, and its simplicity aligns with the grassroots operational environment, making it the most practical and sustainable choice for ensuring fair turn-taking and reducing conflicts.

1.1 Purpose and Description

The main purpose of the **TagSakay** project is to streamline the tricycle queueing system in Barangay Communal by introducing a digital, RFID-based tap system. This solution seeks to automate the recording of tricycle drivers' rotations, enabling real-time updates on a visual display and generating structured digital records for administrative monitoring.

TagSakay will include a tap station powered by an ESP32 microcontroller, equipped with an RFID reader and display screen, and connected to a cloud-ready backend. The data will be transmitted to a web-based dashboard developed using modern frameworks such as Vue.js and Node.js, with reports stored in a PostgreSQL database.

This low-cost solution is designed to minimize manual intervention, ensure queue fairness, and provide the barangay

administration with accurate insights into daily operations. It empowers both drivers and terminal staff by reducing ambiguity, resolving disputes faster, and encouraging transparency through technology.

1.2 Objectives

1.2.1 General Objective

The general objective of this capstone project is to develop a low-cost RFID-based tapping system named "TagSakay" that streamlines the tricycle queue rotation process in Barangay Communal, Davao City, enhancing operational efficiency, accuracy, driver fairness, and administrative transparency through the integration of real-time digital tracking and queue monitoring features.

1.2.2 Specific Objective

1. To design a system that logs driver queue entries using RFID technology, eliminating manual processes such as chalkboard listings and verbal coordination.
2. To enable real-time visual updates of the queue status through a public-facing display, ensuring transparency and accountability in the queueing process.
3. To improve queue processing efficiency during peak hours, holidays, and high-traffic periods through automated queue management and reduced processing time.
4. To automate the generation of daily and weekly reports for route and rotation monitoring, providing centralized digital records and actionable insights

for terminal planning and operational optimization.

5. To develop a dashboard that allows barangay staff or transport officers to track tricycle queue data efficiently, enabling effective validation and audit of queue entries.
6. To reduce administrative disputes by maintaining an accurate and transparent queue record, ensuring fairness in driver rotation order, and minimizing conflicts.

1.3 Scope and Limitations

Scope:

This project is limited to the implementation of a single RFID-based tap-in station designed specifically for the CATICOMODA Route A-29-A tricycle terminal in Barangay Communal. It will accommodate up to 40 registered tricycle drivers, each of whom will be issued an RFID card. The system will include one active portable tap station, a display screen to show real-time queue positions, and a web-based administrative interface to generate reports and monitor activity. The system will be deployed under a canopy-type terminal with a power solution that may include power banks or solar kits, and it will rely on Wi-Fi or mobile data for connectivity.

Limitations:

However, the system is dependent on the availability of a stable power source and internet connection. In the absence of connectivity, real-time synchronization and report generation will be unavailable until reconnection occurs. Moreover, the RFID system assumes that drivers will consistently comply

with the tap-in procedure. The system does not support automatic tap-out logging and lost or damaged RFID cards may require manual intervention. Features such as multi-language support, GPS tracking, and automated facial recognition are not included in this version. Additionally, the system is primarily designed for tricycle queue monitoring and does not address other transport management concerns, such as fare tracking or passenger volume. While the current iteration focuses on RFID due to its specific advantages for this pilot, the project acknowledges the dynamic nature of technology and remains open to integrating or evolving towards other solutions (e.g., vision-based systems for automatic vehicle identification) in future phases based on performance, cost-effectiveness, and community feedback.

II - REVIEW OF RELATED LITERATURE AND SYSTEMS

2.1 Review of Related Literature

Studies from the Philippines provide valuable insights into the local transportation landscape and the applicability of technology-driven solutions for public conveyances like tricycles and jeepneys.

2.1.1: A NodeMCU-Based Contactless Fare Management System for Tricycle Commuters

The implementation of RFID technology in Philippine public transportation has shown promising results through the work of Calis and Aba Jr. [4], who developed an RFID-based contactless fare management system specifically for

tricycle commuters in Negros Oriental, Philippines. Their comprehensive research involved testing with a tricycle driver, ten passengers, and an engineer, utilizing an Arduino/NodeMCU RFID system to modernize payment processes and reduce dependency on cash handling. The findings demonstrated that the system was both feasible and reliable, resulting in smoother fare transactions and overwhelmingly positive feedback from users. This research directly supports TagSakay's focus on tricycle queue management by validating the potential for RFID technology in small-scale public transport applications.

2.1.2: Queuing Management System for Lingayen Jeepney Terminal

Building upon the success of contactless systems, Borje et al. [5] addressed the broader challenge of queue management through their research on a queuing management system for Lingayen Jeepney Terminal in Dagupan City, Philippines. Using Scrum methodology and various data collection tools, they developed a web-based system aimed at improving ticketing, record-keeping, and queuing processes for jeepney drivers and terminal staff. While their solution primarily focused on web-based queue order management rather than physical queue enforcement, the positive user feedback on system performance underscores the critical need for effective queue management tools in the local transport sector and validates the concept of visual displays for queue management systems.

2.1.3: Jeepney Queuing and Tracking System (JEEPQTS)

Expanding the scope of technological solutions, Sotto [6] explored a Jeepney Queuing and Tracking System (JEEPQTS) across various routes in the Philippines. This innovative SMS-based platform enabled drivers and commuters to communicate availability and transportation needs through text messaging, with a centralized system providing real-time updates to all participants. Although the system demonstrated improvements in matching efficiency and reduced idle time for drivers, it did not address physical queue order management directly. This limitation reveals a significant gap in real-time, on-ground queue enforcement that TagSakay aims to fill through its comprehensive approach to physical queue management.

2.1.4: Internet of Things-Based Revenue Collection System for Tricycle Vehicle Operators

The integration of Internet of Things (IoT) technology in public transportation has been demonstrated by Maduka et al. [7], who developed an "Internet of Things-Based Revenue Collection System for Tricycle Vehicle Operators" in Akwa Ibom State, Nigeria. Their comprehensive system utilized Android applications and microcontroller circuits, specifically aimed at reducing revenue leakages in tricycle operations. While broader in scope than queue management alone, this research highlights how digital systems can effectively manage and collect revenue from tricycle drivers, demonstrating the potential for comprehensive technological solutions in similar public transport contexts and supporting TagSakay's integrated

approach to transportation management.

2.2 Review of Related Systems

International studies and system implementations offer a broader perspective on RFID and IoT applications in transport, addressing various aspects from fare collection to smart logistics, and demonstrating solutions that can be supportive of TagSakay's goals.

2.2.1: Bus Fare Collection System Using RFID and GPS

Zaman et al. [8] developed a comprehensive "Bus Fare Collection System Using RFID and GPS" in Dhaka, Bangladesh, creating an Automatic Fare Collection (AFC) system that aimed to streamline ticketing by replacing paper tickets, reducing processing time, and avoiding fare disputes. The system automatically calculated and deducted fares based on travel distance while recording passenger locations for operational analysis. Despite demonstrating improvements in fare collection and offering discounted fares to users, the system revealed important limitations, including an inability to handle new users and GPS accuracy issues in indoor environments. This research directly supports TagSakay's concept of automated tracking, highlighting the critical importance of offline capabilities for outdoor queue monitoring, given the identified GPS limitations.

2.2.2: Automated Public Transport Fare Collection System Using RFID

In Mumbai, India, Joshi et al. [9] demonstrated an "Automated Public Transport Fare Collection

System Using RFID" using a Raspberry Pi-based prototype. Their system successfully streamlined payment processes and reduced human contact while integrating RFID technology for simplified commuting experiences. Additionally, the system generated valuable operational data for transport authorities, reinforcing the efficiency of RFID in fare collection and demonstrating its capacity to provide comprehensive admin-side reporting capabilities that can inform operational decisions.

2.2.3: Design of Smart Bus Fare Collection System Using RFID

Ramani et al. [10] designed a "Smart Bus Fare Collection System Using RFID" in Visakhapatnam, India, creating a comprehensive prototype built with Arduino UNO, RFID reader, GSM module, and LCD display components. Their implementation featured integrated software for fare deduction via microcontroller, highlighting the effectiveness of hardware-software integrated solutions for automated fare collection. This research demonstrates how different hardware components can be successfully integrated to create a functional automated fare collection system.

2.2.4: RFID-Based Bus Tracking System

Omkar Aher et al. [11] proposed an "RFID-Based Bus Tracking System" in Pune, India, featuring an innovative architecture that used RFID tags on buses and readers at stops connected via IoT infrastructure. The system aimed to provide real-time bus location information and estimated time of arrival (ETA) data to passengers and

operators. Their findings suggested improved transit efficiency and optimized scheduling capabilities for transport operators. While focused on vehicle tracking rather than queue management, the use of RFID at fixed points for data collection is highly relevant to TagSakay's approach to monitoring queue rotations and driver positions.

2.2.5: Conductor-less Bus Ticketing System Using RFID Technology

Agarwal et al. [12] developed a "Conductor-less Bus Ticketing System Using RFID Technology" in India, creating an Arduino-based contactless fare system that demonstrated significant time and cost savings for operators through automatic ticket generation and fare deduction processes. This research emphasizes the operational benefits and increased security provided by automated systems, demonstrating how technology can reduce labor costs while improving service reliability and reducing opportunities for revenue leakage.

2.2.6: Hospital Queue Wait Time Prediction

Mishra et al. [13] proposed a multi-stage queue model using RFID tag-sensors for hospital queue wait time prediction in Navi Mumbai, India. Their IoT queue-management system aimed to reduce patient anxiety and improve hospital efficiency through better queue management and wait time estimation. Although implemented in healthcare rather than transportation, the application of RFID for real-time queue monitoring and data collection is directly analogous to TagSakay's goals for physical queue management, demonstrating the

versatility of RFID technology across different domains.

2.2.7: Smart Ticketing for Academic Campus Shuttle Transportation System Based on RFID

Abayomi-Alli et al. [14] implemented a "Smart Ticketing for Academic Campus Shuttle Transportation System Based on RFID" in Ota, Nigeria, creating a simple, low-cost system that utilized RFID student cards to effectively eliminate paper tickets and revenue leakage. The successful implementation of this cost-effective RFID solution on a campus shuttle system provides a scalable model for TagSakay, particularly demonstrating the feasibility of deployment in resource-constrained environments where cost-effectiveness is a primary concern.

2.2.8: Integrated Ticketing System for Public Transport Using RFID Technology

Senthazhai et al. [15] proposed an "Integrated Ticketing System for Public Transport Using RFID Technology" in Tamil Nadu, India, utilizing a unified "Smart Master Card" that integrated RFID and IoT technologies for multi-modal transit applications. This comprehensive system aimed to eliminate manual fare collection, reduce operational costs, and improve efficiency through centralized payment management across multiple transport modes. The research highlights the potential for broader integration and centralized management of transport data, supporting TagSakay's vision for comprehensive transportation management.

2.2.9: Fare Transfer in Public Transport Using RFID

Singh et al. [16] designed an "RFID tap-to-pay machine for auto rickshaws and buses" in India, achieving remarkable cashless transactions within two seconds of card presentation. This efficient system instantly transferred fare amounts to the driver's account, emphasizing faster transaction processing and the convenience of cashless systems for both passengers and operators. These aspects of rapid processing and immediate fund transfer can be extended to automated data collection and record-keeping for TagSakay, bolstering fair practice when it comes to transparency

2.2.10: A Review of Passenger Counting in Public Transport: Concepts with Solution Proposal

Radovan et al. [17] conducted "A Review of Passenger Counting in Public Transport: Concepts with Solution Proposal," providing a comprehensive international literature review of passenger counting methods and technologies. They noted that RFID technology offers fast, contactless passenger counting capabilities but requires every passenger to carry a tag and may miss fare evaders who bypass the system. This review highlights both the significant advantages and inherent limitations of RFID for counting and monitoring applications, informing TagSakay's development regarding potential challenges in tag management and verification processes to prevent system misuse.

2.2.11: Ticket Generating and Bus Crowd Controlling Smart Bus Stop System Using IoT

Priya et al. [18] proposed a "Ticket Generating and Bus Crowd Controlling Smart Bus Stop System Using IoT" in Tamil Nadu, India, using RFID cards, Arduino, and GSM modules to create a comprehensive passenger management system. This innovative system automatically generated tickets via RFID tags and notified passengers via SMS, effectively reducing bus crowding and improving passenger flow management. The concept of using RFID for automated ticketing and crowd control is highly relevant to queue management applications and supports TagSakay's objectives for automated queue processing.

2.2.12: Automatic Bus Fare Collection System (undergraduate project report)

Donnel J et al. [19] presented an "Automatic Bus Fare Collection System" as an undergraduate project in India, creating an Arduino UNO-based fare machine with an RFID card top-up portal. Their system demonstrated how RFID systems can eliminate the need for cash transactions and facilitate online recharging capabilities for users. This research indicates the feasibility of comprehensive RFID-based payment solutions that can be integrated into broader transportation management systems, supporting TagSakay's vision for automated operations.

2.2.13: AUTOMATED CROWD MANAGEMENT IN PUBLIC TRANSPORT

Jadhav et al. [20] demonstrated an "Automated Crowd Management in Public Transport" system in Bengaluru, India, using RFID smart-cards, ARM controllers,

NodeMCU, and IR sensors to create a comprehensive passenger management solution. Their embedded system provided real-time seat monitoring and automated fare/messaging capabilities, showing an efficient crowd-management solution that significantly reduces human error in transport management. This research aligns with TagSakay's efforts to minimize manual processes and improve operational efficiency through automation.

2.2.14: Design and Implementation of a Low-Cost Universal RFID Wireless Logistics Terminal in the Process of Logistics Traceability

Yu et al. [21] designed and implemented a "Low-Cost Universal RFID Wireless Logistics Terminal" in China, creating an RFID terminal integrated with LoRa, RJ45, and Wi-Fi connectivity options. Their system achieved high accuracy and real-time logistics traceability across different communication protocols. While applied in logistics rather than transportation, this research emphasizes the reliability and real-time capabilities of RFID in tracking applications, which is directly relevant to TagSakay's data collection requirements and demonstrate the potential for multi-protocol connectivity.

2.2.15: Smart Trolley with Automatic Billing System using RFID

Dangat et al. [22] developed a "Smart Trolley with Automatic Billing System using RFID" in Pune, India, creating a prototype that successfully automated item scanning and real-time billing processes, significantly reducing checkout time and improving shopping efficiency. This demonstrates RFID's broader

application in real-time data capture and automated processes across various domains. Though not specifically for transportation, this system shows how RFID technology can improve process efficiency and reduce manual work, supporting the principles underlying TagSakay's automated approach to queue management.

Research Gap

Despite the range of studies and systems reviewed, several key gaps remain unaddressed, particularly in the context of small-scale, community-level transportation like that in Barangay Communal. Most existing systems did not specifically focus on ensuring fair turn-taking among tricycle drivers, which is a crucial aspect in maintaining order and avoiding disputes in local terminals. Additionally, there is a noticeable absence of real-time visual displays that show the current queue order—an essential feature for transparency and efficient driver coordination.

Many of the existing solutions were also designed for more resource-rich settings and did not consider the constraints of low-cost environments. These include areas with limited access to stable electricity, internet connectivity, or advanced infrastructure—conditions commonly found in rural and barangay-level terminals. Lastly, a significant gap was the lack of deployment and testing within actual local communities, which limits the practical validation of these systems in real-world scenarios. This highlights the need for solutions like TagSakay that are not only technically sound but also tailored for and tested within grassroots settings.

III - TECHNICAL BACKGROUND

Supporting Theories and Tools:

- **RFID Technology:** Helps identify drivers through cards and scanners.
- **ESP32-S3:** Small, low-powered, but affordable microcontrollers used in many DIY electronics and smart systems.
- **Vue.js and Workers Wrangler:** Web development tools used to create dashboards and real-time interfaces.
- **PostgreSQL:** A reliable database used in many systems to safely store data.
- **Internet of Things (IoT):** The concept of connecting devices to share and collect data is used in many modern automation projects.

System Overview

The TagSakay system is designed to improve the way tricycle queues are managed in Barangay Communal, Davao City, by replacing manual processes with a low-cost, automated RFID-based solution. The system provides a reliable and transparent way to track driver turns, reduce disputes, and support local transport officers in managing terminal operations more efficiently. Its architecture is divided into three main components: Input, Process, and Output, each working together to transform RFID taps into real-time queue data and administrative reports.

Input

The system starts when tricycle drivers tap their RFID cards on the tap station upon arriving at the terminal. Each card is assigned to a specific driver and acts as a digital signature to mark their presence in the queue. A manual keypad is also included, allowing the terminal administrator to override the system if a card is lost, a tap is missed, or if the queue needs adjustment. These two input methods help ensure accurate and flexible recording of queue entries.

Process

Once a driver taps their card, the main ESP32 controller reads the RFID tag and immediately displays the driver's number on the terminal LED Matrix screen. It also plays a confirmation beep for audio feedback. If the internet is temporarily unavailable, the data is stored locally in the memory or SD Card. When connected, the data is sent via Wi-Fi or mobile hotspot to a CloudFlare Worker, which handles the processing and saves the information into a PostgreSQL database. The system also powers a Vue.js-based web dashboard, where staff can view live queue activity, generate logs, and access real-time statistics. This flow ensures that both live and historical data are accurate, consistent, and easy to manage.

Output

The final part of the system delivers real-time and historical queue information through various interfaces. A live queue display on the terminal shows the current list of drivers, including those who have tapped in and who are next in line. Audio feedback confirms each tap to drivers

without needing to look at the screen. On the staff side, the admin dashboard provides access to detailed reports, tap logs, and system controls. These outputs make it easy to monitor activity, resolve disputes, and ensure fairness in driver rotations.

CHAPTER 4 - SYSTEM METHODOLOGY AND DESIGN

4.1 Methodology

The TagSakay RFID Queue Monitoring System was developed using a **Modified Agile** methodology tailored for hardware-software co-development. Weekly sprint cycles enabled rapid iteration on both firmware and backend architecture while addressing field-specific challenges.

Sprint Cycles Overview

Table 1.0 Sprint Cycles and Deliverables for Hardware-Software Co-development

Weeks	Deliverables
1 – 2	Backend and firmware foundation: Cloudflare Workers setup (Hono), Neon PostgreSQL schema (Drizzle ORM), primary ESP32 hardware initialization
3 – 4	RFID backend processing, authentication, and queue state model

5	OWASP-aligned security implementation
6 – 7	Vue.js admin dashboard UI and REST integrations
8	Communication model update: Migrated from WebSockets → HTTP Polling to avoid Durable Objects billing
9 – 10	Dual-ESP32 firmware with a standalone LED Matrix controller, modular architecture
11 – 12	Integration, stress testing, and deployment documentation

Stakeholder Collaboration

Stakeholder participation was integrated into the sprint process:

Table 1.1 Stakeholder Focus Groups and Contributions to System Requirements

Stakeholder Group	Focus Contribution
Tricycle Drivers	Queue fairness, RFID tap workflows
Barangay Officials	Administrative controls, reporting requirements

Technical Advisers	Hardware-network architecture reviews
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Hardware–Software Integration Strategy

The integration between hardware and software focused primarily on ensuring that the system could operate efficiently and reliably under real-world conditions. Priority was placed on achieving accurate and fast RFID scanning to avoid delays in queue processing and to maintain system responsiveness. Synchronization between the main ESP32 device and the LED display controller was also emphasized to ensure that queue updates remain consistent regardless of network status.

Given the outdoor deployment environment, the display system was optimized for high visibility under direct sunlight, while the firmware was reinforced to continue functioning even during intermittent network connectivity. Finally, power management techniques were applied to support long-duration operation using portable or limited solar power sources, making the system suitable for barangay terminals where stable electrical access may not always be guaranteed.

4.2 Data Gathering

Multiple research techniques were employed to accurately establish the system requirements and ensure that the solution addresses the actual challenges faced in tricycle queue management. Structured interviews with tricycle drivers were conducted to

gather insights on existing queue practices, specifically focusing on operational pain points, fairness of rotation, and expectations for RFID responsiveness and usability.

In addition to interviews, a survey questionnaire was distributed to a wider group of drivers to quantify common concerns, validate the importance of identified issues, and collect feedback on preferred system functionality. The survey confirmed that most drivers experienced delays and disputes caused by unclear queue order, and that they strongly preferred real-time visibility of their position in the rotation.

Field observations at operational terminals complemented the interview and survey findings by revealing environmental considerations and logistical constraints. Observed issues included peak-hour congestion leading to unintentional queue skipping, human errors in manual recording, exposure to dust, heat, and glare that could affect device durability and display readability, and unreliable access to power outlets that demanded power-efficient hardware. Insights from barangay officials and IT consultants contributed additional administrative requirements, including the need for queue override controls, history logging, and monitoring capabilities to ensure transparency and accountability in queue rotation. Collectively, these findings drove key design decisions such as FIFO-based automated queue progression, fast feedback through audio and visual cues, offline operational capability, and hardware suited for outdoor deployment.

4.3 Requirements Analysis

The requirements analysis phase systematically identified and documented the functional, non-functional, and technical requirements for the TagSakay system based on data gathered from stakeholders, site observations, and technical assessments. This comprehensive analysis ensures that the developed system addresses real user needs while maintaining technical feasibility and operational viability.

4.3.1 System Requirements

4.3.1.1 Functional Requirements

RFID Tap Detection: The system must detect an RFID tag tap within at most one second, using debounce logic to prevent duplicate scans.

Tag Validation: The system must validate whether a scanned RFID tag belongs to a registered driver or requires handling as an unregistered tap.

FIFO Queue Rotation: The system must maintain a First-In-First-Out queue structure, automatically advancing positions when a driver boards.

HTTP Command Polling: The system must regularly poll the backend every five seconds to obtain updated commands and queue state synchronization data.

Admin Overrides: The system must allow administrators to manually adjust or override queue positions through both the keypad and the web dashboard.

Registration Mode: The system must support switching into a dedicated registration mode for adding or

assigning RFID tags without affecting the operational queue.

Device Heartbeat: The system must periodically report its operational status to the backend for monitoring purposes.

Offline Queue Continuity: The system must continue managing and updating queue operations locally during network outages and automatically synchronize once reconnected.

4.3.1.2 Non-Functional Requirements

Offline Mode: The system must operate for a minimum of eight hours without a stable power or network source.

Low Power Operation: The system must optimize power usage by adjusting LED panel brightness and minimizing unnecessary processing.

Tap Response Time: The system must provide feedback to the user in under one second after an RFID tag is tapped.

Display Readability: The system must ensure the queue display remains clearly visible under outdoor lighting and sunlight exposure.

Secure Access: The system must enforce secure authentication using role-based access control and JWT mechanisms for all dashboard interactions.

Network Resilience: The system must continue functioning properly during intermittent connectivity by applying retry and exponential backoff logic.

Data Persistence: The system must preserve queue state and associated records during power interruption or unexpected restarts.

Multi-Device Support: The system must support simultaneous polling and operation of multiple ESP32 devices without performance degradation.

4.3.2 Technical Requirements

4.3.2.1 Software Components

Table 1.2 Software Component Specifications and Development Frameworks

Component	Specification
Cloudflare Workers	Backend runtime executing the REST API for RFID operations and queue processing.
Hono Framework	Provides routing and middleware for request handling and authentication.
PostgreSQL + Drizzle ORM	Stores system data with type-safe queries, schema management, and migrations.

Vue 3 Dashboard	Frontend interface for queue monitoring, device management, and administrative controls.
Custom Rendering Engine	Framebuffer-based UI on the ESP32 display, built without LVGL for efficiency.
HTTP Polling	Periodic device-to-server requests to fetch commands and queue updates.
RBAC Security Stack	Role-based access using JWT, session controls, and rate limiting.

*Note: HTTP polling replaced WebSockets to remove dependency on **paid Durable Objects**, ensuring an almost zero-cost backend operation.*

4.3.2.2 Hardware Components

Table 1.3 Hardware System Components and Microcontroller Specifications

Component	Specification
ESP32 (Main)	Handles RFID scanning, queue logic, TFT rendering, HTTP polling, and UART transmission.
ESP32 (LED Controller)	Controls the LED matrix display and manages SD card access for lookup and logs.
PN532 Reader	Reads NFC/RFID tags used for tricycle identification.
P5 LED Matrix (x2)	Outdoor-facing passenger display for queue numbers and status.
TFT Display (480×320)	Local on-device interface rendered through a custom framebuffer UI.
Keypad + Buzzer	Provides admin input and audible feedback during operations.
SPIFFS	Stores cached data and fallback resources on the main ESP32.
SD Card Module	Generates logs and stores reference data for LED display operations.

UART Serial Link	Communication channel between the main ESP32 and the LED controller.
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4.3.2.3 Client Device Requirements (Dashboard Access)

Table 1.4 Minimum Client Device Requirements for Admin Dashboard Access

Requirement Category	Specification
Device Type	Desktop, laptop, or mobile device capable of running a modern web browser.
Operating System	Windows 10/11, macOS, Linux, Android 10+, or iOS 13+.
Web Browser	Latest versions of Chrome, Firefox, Edge, or Safari.
Internet Connection	Stable connection (at least 5 Mbps) for real-time dashboard updates.

Screen Resolution	Minimum 1280×720; recommended 1920×1080 for full dashboard visibility.
Hardware Specs	At least dual-core processor and 4 GB RAM for smooth rendering.
JavaScript Enabled	Required for Vue 3 components and interactive dashboard features.
Cookies/Storage	Must allow session cookies for authentication and API access.

4.4 System Design

The system design phase translates the analyzed requirements into concrete architectural models, diagrams, and specifications that guide the development and implementation of TagSakay. This section presents the conceptual framework, use case interactions, data flow architecture, and process flows that define how the system operates.

4.4.1 Conceptual Framework

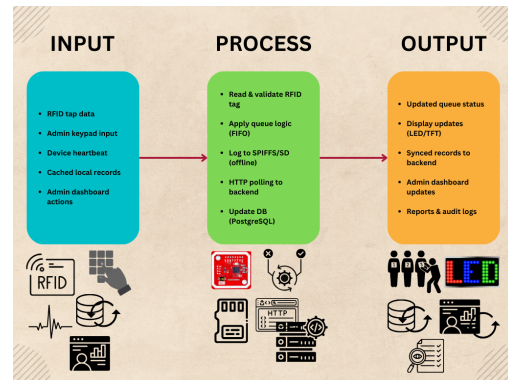


Figure 1.0 Conceptual Framework (See Full Image in Appendix A)

4.4.2 Use Case Diagram

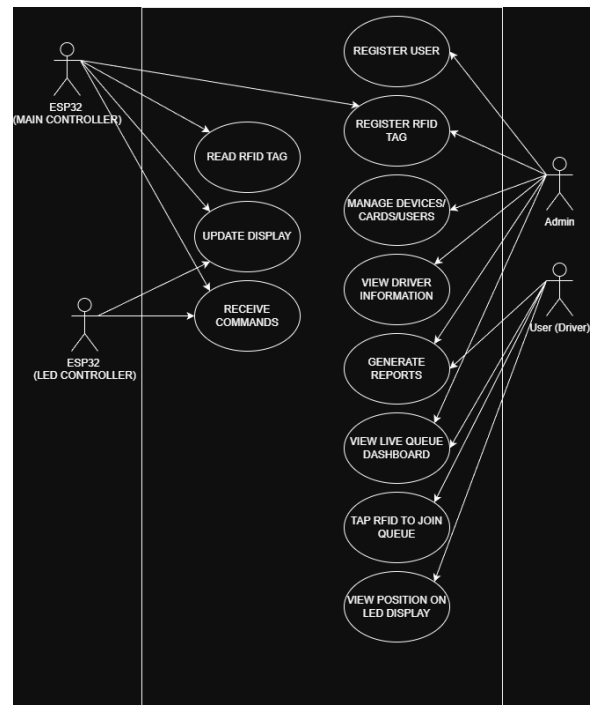


Figure 1.1 Use Case Diagram (See Full Image in Appendix A)

4.4.3 Context Flow Diagram

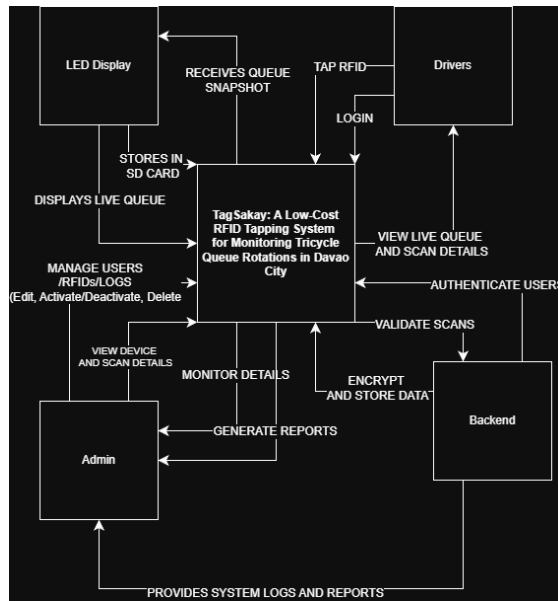


Figure 1.2 Context Flow Diagram
(See Full Image in Appendix A)

4.4.4 Data Flow Diagrams

DFD 0

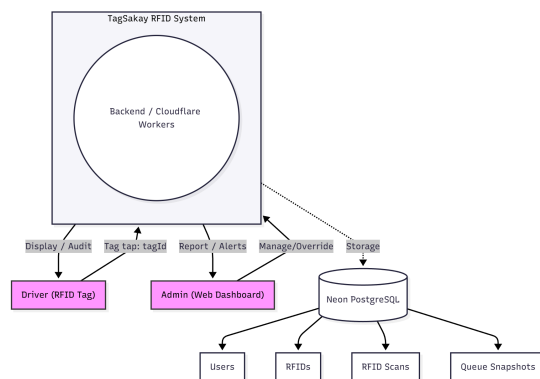


Figure 1.3 Data Flow Diagram 0 (See Full Image in Appendix A)

DFD 1

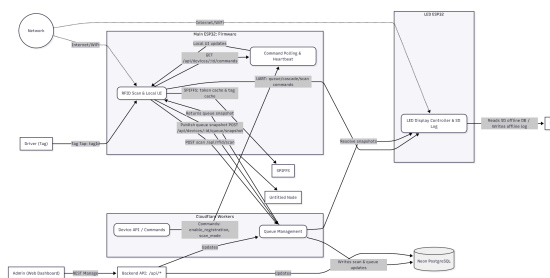


Figure 1.4 Data Flow Diagram 1 (See Full Image in Appendix A)

DFD 2

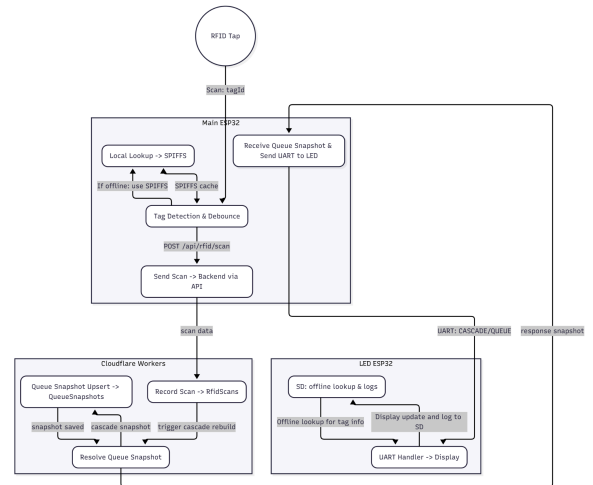


Figure 1.5 Data Flow Diagram 2 (See Full Image in Appendix A)

4.4.5 System Flowchart

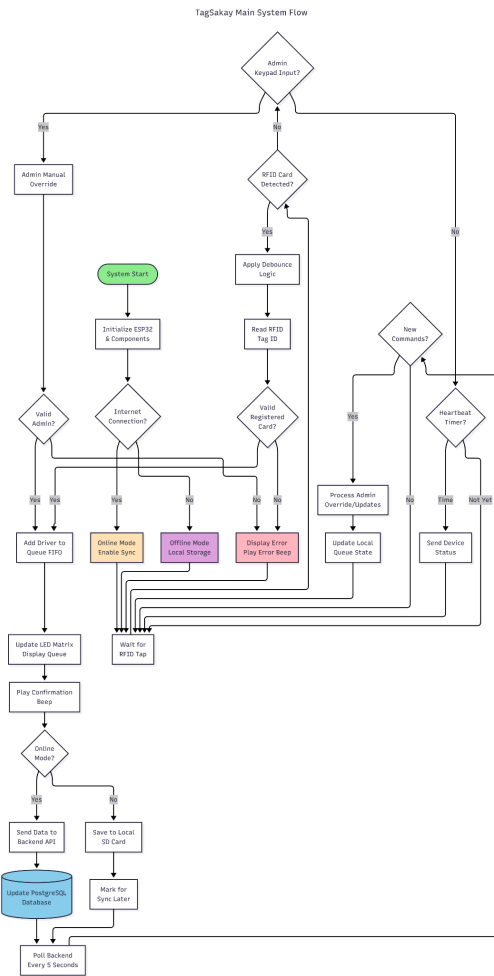


Figure 1.6 System Flowchart - Firmware (See Full Image in Appendix A)

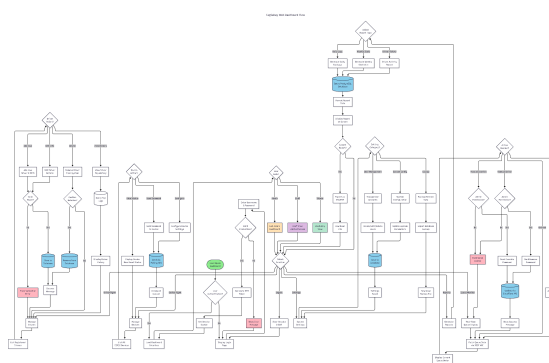


Figure 1.7 System Flowchart - Webview (See Full Image in Appendix A)

4.4.6 System Architecture

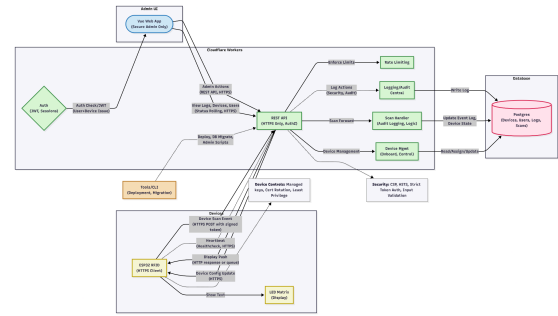


Figure 1.8 System Architecture (See Full Image in Appendix A)

4.5 Development

This chapter outlines the implementation details of the TagSakay system across its three major subsystems: the backend Cloudflare Workers API, the Vue-based administrative console, and the dual-ESP32 firmware and LED display hardware stack. Each subsection describes the technologies used, development decisions made, and workflows followed during system construction.

4.5.1 Backend Workers Implementation

The backend was implemented under the backend-workers/ directory using a serverless, low-cost, and high-availability architecture.

Table 1.5 Backend Workers Implementation and Feature-based Route Structure

Component	Description
Technology Stack	Cloudflare Workers, Hono framework, Drizzle ORM, Neon PostgreSQL
Route Structure	Feature-based routers under <code>src/routes</code> such as <code>auth.ts</code> , <code>rfid.ts</code> , <code>device.ts</code> , and <code>user.ts</code> ;

	all responses follow the { success , message , data } format
Database Layer	Schema defined in src/db/schema.ts for users, devices, RFID tags, scans, and audit logs; Drizzle migrations under backend-workers/drizzle/
Security Middleware	JWT bearer verification, API key enforcement, PBKDF2-SHA256 (600k iterations), full OWASP claim set, and rate-limiting tiers (Auth: 5/min; API: 100/min; Device Registration: 3/hr)
RFID Processing	rfid.ts handles scan ingestion, duplicate detection, and offline-buffer replay logic
Device Lifecycle	device.ts manages registration, heartbeat intake, and remote mode switches; telemetry is stored for dashboard monitoring
Operational Tooling	Database cleaning, migration, and seeding scripts (migrate.ts , clean-db.ts , seed.ts); .dev.vars for secrets; Wrangler for deployment consistency

4.5.2 Frontend Admin Console

The administrative dashboard was implemented under frontend/ using Vue 3 and TypeScript.

Table 1.6 Frontend Admin Console Implementation and Feature Modules

Component	Description
Technology Stack	Vue 3, TypeScript, Vite, Tailwind CSS

Routing & Authorization	Role-guarded routing (SuperAdmin, Admin, Driver); login and registration validation aligned with backend logic
Feature Modules	Dashboard KPIs, RFID/tag management, device monitoring, queue visualization; reusable UI components (dialogs, forms, toasts)
API Integration	Implemented via src/services/api.ts with automatic JWT insertion and credentials: "include" for session cookies
Build & Deployment	Vite build outputs are deployed to Cloudflare Pages; VITE_API_URL toggles between local and production environments
Security Alignment	Client-side enforcement of RFID format, input sanitization, and password-policy checks to reduce invalid backend traffic

4.5.3 Firmware and LED Display Stack

Firmware development was conducted in the TagSakay_Fixed_Complete/ directory, targeting a dual-ESP32 architecture (main controller + LED matrix controller).

Table 1.7 Firmware Modules and Functional Responsibilities for ESP32 Controllers

Module	Function
--------	----------

HTTPPolling.cpp	Polls backend for commands (every 5 seconds) and heartbeats (every 30 seconds)
ApiModule.cpp	Handles REST requests, exponential backoff, and offline buffering using SPIFFS/SD
RFIDModule.cpp	Reads PN532 RFID tags with debounce and UID validation
DisplayModule.cpp	Renders TFT UI (480×320) with custom framebuffer renderer
UARTModule.cpp	Transmits queue state to LED matrix controller via UART
KeypadModule.cpp	Admin override input (registration mode, rewind, maintenance); syncs via HTTP command channel

Configuration & Resilience

Table 1.8 System Configuration and Resilience Measures for Offline Operation

Area	Description
Configuration	<code>Config.h</code> stores API base URL, device MAC identity, polling intervals, and feature flags
Offline Operation	Buffered scans stored in SPIFFS/SD; auto-replay to <code>/api/rfid/scan</code> once reconnected
Watchdogs	Prevent firmware lockups during brownouts; safe-mode keypad combination for recovery

LED Controller	Dedicated ESP32 for P5 LED matrices; receives queue payloads from main ESP32
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4.5.4 Integration & Deployment Workflow

The project follows a structured deployment process to ensure consistency across backend, frontend, and firmware components.

Table 1.9 Step-by-Step Integration and Deployment Workflow

Step	Procedure
1. Backend Setup	<code>npm install</code> → configure <code>.dev.vars</code> → run <code>npm run db:setup</code> → start via <code>npm run dev</code> ; production deployed via <code>npm run deploy</code> with Wrangler secrets
2. Frontend Build	Local development via <code>npm run dev</code> ; production built with <code>npm run build</code> and deployed to Cloudflare Pages
3. Firmware Flashing	Update <code>Config.h</code> with API URL & device key → compile & flash to both ESP32s
4. Device Registration	Register device in the admin dashboard, assign location, generate API key; heartbeat verifies device status
5. Monitoring & Observability	Audit logs, queue metrics, and error rates tracked through Neon dashboards and Drizzle Studio

4.5.5 Configuration Management & Security Hardening

Robust configuration and security policies were applied across all system layers.

Table 1.10 Security Hardening Policies and Configuration Management Procedures

Area	Description
Secrets Handling	<code>.dev.vars</code> for local; Cloudflare Secrets for production; JWT/session secrets generated via secure random bytes
Rate Limiting	Tiered controls per route; lockout counters with clean <code>429</code> responses and <code>retryAfter</code> hints
Input Validation	Shared regex rules for RFID UUIDs, MAC addresses, and email formats (RFC 5321 compliance)
Observability	Logs captured for 15+ security event types (LOW–CRITICAL); persisted for audit and incident response

4.6 System Testing and Validation

This section consolidates the verification activities conducted on the TagSakay platform and demonstrates how each subsystem satisfied the functional and non-functional requirements defined in Section 4.3.

4.6.1 Testing Strategy

Testing followed a layered approach so that defects could be isolated early and retested as subsystems matured.

Firmware and backend modules were validated first through automated scripts, followed by API-level integration exercises, and capped by stakeholder-facing acceptance tests in the Barangay Communal terminal.

4.6.1.1 User Acceptance Testing

User Acceptance Testing (UAT) was carried out with three respondent groups: tricycle drivers, barangay personnel, and IT personnel. Each group evaluated critical system features using a five-point Likert scale, where 1 represented strongly disagree and 5 represented strongly agree. As shown in Table 1, all features achieved mean scores above 4.0, indicating strong perceived functionality across stakeholder groups and validating the user-facing requirements.

Table 1.11 Descriptive Statistics for Perceived System Functionality by Respondent Group

System Feature	Drivers Mean	IT Personnel Mean	Barangay Personnel Mean
RFID Tag Scanning	4.54	5.00	5.00
Queue Tracking	4.71	4.00	4.00

Tag Registr-ation	4.46	5.00	5.00
Queue Monito-ring	4.60	5.00	4.00
Device Monito-ring	4.49	5.00	4.50
Data Loggin-g	4.37	5.00	5.00
Report Gener-a-tion	4.46	5.00	4.50
Authen-tication	4.46	5.00	5.00

4.6.1.2 Unit Testing (per module)

Software unit tests were implemented for critical backend utilities (authentication helpers, RFID validation, rate limit counters) using Vitest, while the frontend relied on Vue Test Utils to verify form validation, API wrappers, and state management. Firmware modules were bench tested individually; for example, `RFIDModule.cpp` was subjected to repeated UID reads under varying power levels, and `HTTPOlling.cpp` was exercised with mocked HTTP

responses to confirm retry spacing and exponential backoff logic.

4.6.1.3 Integration Testing (API and system communication)

Integration testing focused on the contract between ESP32 devices and the Cloudflare Workers API. Scripts under `scripts/` replayed tap events, simulated packet loss, and validated that queued scans were accepted only once. Parallel tests confirmed that the Vue dashboard consumed the same API responses without schema drift, ensuring that firmware, backend, and frontend interpreted queue payloads consistently.

4.6.1.4 System Testing (end-to-end)

End-to-end tests reproduced the full queueing workflow in the Barangay Communal terminal. Test drivers tapped RFID cards, observed LED matrix updates, and completed dispatch confirmations while barangay personnel monitored the admin dashboard. These trials validated FIFO queue enforcement, visual and audio feedback, report generation, and recovery behavior when Wi-Fi connectivity was intentionally interrupted.

4.6.2 Test Cases and Results

Representative manual and automated test cases are summarized in Table 2. Each case documents the expected and actual outcomes gathered from logs, dashboard screenshots, and serial traces.

Table 1.12 Summary of Core Test Cases

Test ID	Scenario	Expected Result	Actual Result	Status
TC-001	RFID tap during normal operations	Tap recorded within 1 second; driver moves to the tail of the FIFO queue	Tap acknowledged in 730 ms; LED display updated, and the backend stored the transaction	Pass
TC-002	Duplicate tap within debounce window	Second tap ignored; no duplicate scan saved	Firmware rejected duplicate; no additional scan row created	Pass
TC-003	Offline buffer replay after 10-minute outage	Locally cached scans synchronize once the network returns	Queue replayed in timestamp order; dashboard showed continuous rotation	Pass
TC-004	Admin override via keypad	Manual adjustment reflected on the LED display and dashboard audit log	Override applied; audit log captured user, timestamp, and change reason	Pass
TC-005	Login rate limit exceeded	Account locked for 15 minutes with countdown UI feedback	Lockout triggered on 5th failure; <code>retryAfter</code> rendered on login page	Pass

TC-006	Report generation for the previous week	CSV exports contain all scans with driver metadata	Generated CSV matched Neon query; users verified totals during review session	Pass
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4.6.3 Web Integration Testing

Web integration testing ensured that the Vue admin console remained synchronized with the Cloudflare Workers API. Manual browser walkthroughs (Chrome, Edge, and Android WebView) navigated the dashboard, triggered RFID scans via mock API payloads, and verified that the WebSocket-to-HTTP polling migration did not introduce stale data. Additional browser-based tests covered mixed authentication states (session cookie only, bearer token only, combined). They confirmed that the UI honored role-based guards when switching between SuperAdmin, Admin, and Driver views on both desktop and mobile breakpoints.

4.6.4 Security Testing

Security validation combined automated scanning and manual penetration tests:

Table 1.13 Security Tests and Detailed Validation

Authentication hardening	scripts/test-login.js executed rapid login attempts to guarantee the PBKDF2 hashing (600k iterations) and lockout policy resisted brute-force attempts.
Rate limit enforcement	OWASP-aligned buckets (auth: 5 per minute, API: 100 per

	minute, device registration: 3 per hour) were exercised through Postman collections with ramp-up traffic to validate deterministic 429 responses and retryAfter hints.
JWT and session review	Issued tokens were decoded to confirm full claim sets (iss, aud, sub, iat, exp, nbf, jti) and four-hour expirations. Session cookies were inspected for HTTP-only and Secure flags.
Static analysis	npm run lint and dependency checks ensured no known vulnerabilities in backend or frontend dependencies, and firmware builds were scanned for unsecured credential storage.

4.6.5 Error Handling and Troubleshooting

Operational readiness relies on layered monitoring. Firmware serial logs flag RFID read errors, queue overflows, and watchdog resets, while the backend emits structured JSON logs for authentication failures, device heartbeats, and scan ingestion anomalies.

During field deployment, the researcher followed a troubleshooting playbook that maps symptoms to recovery actions (e.g., reissuing API keys, clearing SPIFFS buffers, restarting LED controllers). Combined with the admin dashboard's telemetry

widgets, these measures allowed rapid isolation of faults and minimized downtime during pilot operations.

CHAPTER 5 – RESULTS AND DISCUSSION

This chapter presents and discusses the results of the user testing conducted for the TagSakay: A Low-Cost RFID Tapping System for Monitoring Tricycle Queue Rotations. Survey data were collected after system testing to evaluate both functional and non-functional requirements. Respondents included tricycle drivers, barangay personnel, and IT personnel who interacted with the RFID-based queue system and its web dashboard. The primary purpose of the evaluation was to assess whether TagSakay met its objectives of improving queue transparency, operational efficiency, and monitoring accuracy.

WEIGHT / SCALE

Weight	Range of Mean	Description
5	4.51 – 5.00	Excellent
4	3.51 – 4.50	Very High
3	2.51 – 3.50	High
2	1.51 – 2.50	Fair
1	1.00 – 1.50	Poor

Table 2.1 TagSakay Rating Scale for Functional and Non-Functional Requirements

Table 2.1 presents the five-point Likert scale used during user testing. This scale measured respondents' level of agreement with system performance indicators, where 5=Excellent, 4=Very High, 3=High, 2=Fair, and 1=Poor.

Descriptive Statistics

Accessibility and Ease of Use

Indicator	Mean	Description
The TagSakay system interface is user-friendly and easy to navigate.	4.48	Very High
The system is easy to understand even for first-time users.	4.42	Very High

Table 2.2 Accessibility and Ease of Use Evaluation

Table 2.2 shows that users found the TagSakay system easy to use and understand. The Very High ratings indicate that minimal training is required for effective system adoption.

Queue Management and RFID Processing

Indicator	Mean	Description
Drivers are correctly detected and recorded through RFID scanning.	4.55	Excellent
Queue positions update accurately after each scan.	4.52	Excellent
Duplicate or invalid scans are prevented by the system.	4.47	Very High

Table 2.3 Queue Management and RFID Processing Evaluation

Table 2.3 indicates strong system reliability in detecting RFID tags and maintaining accurate queue order. The Excellent ratings demonstrate the effectiveness of RFID automation in eliminating manual queue handling errors.

System Performance

Indicator	Mean	Description
The system responds	4.44	Very High

quickly to RFID scans.		
Dashboard updates reflect queue changes in real time.	4.46	Very High

Table 2.4 System Performance Evaluation

The results in Table 2.4 show that TagSakay performs efficiently in real-time operations. Users observed timely responses between RFID scans and dashboard updates, supporting operational reliability.

System Availability

Indicator	Mean	Description
The system is consistently available during daily queue operations.	4.45	Very High

Table 2.5 System Availability Evaluation

The Very High rating indicates that the system remains accessible throughout normal operating hours, which is essential for uninterrupted queue monitoring.

Scalability and Performance

Indicator	Mean	Description
The system remains responsive during peak hours.	4.41	Very High

Table 2.6 Scalability and Performance Evaluation

Table 2.6 suggests that TagSakay maintains stable performance even during periods of increased system usage, such as peak tricycle queue times.

User Experience

Indicator	Mean	Description
Overall experience using the TagSakay system as a driver is satisfactory.	4.50	Very High

Table 2.7 User Experience Evaluation

The results indicate high user satisfaction, reflecting the system's practicality and effectiveness in real-world queue operations.

Administrator-Side Evaluation

Accessibility and Monitoring

Indicator	Mean	Description
Administrators can easily access and monitor queue data.	4.46	Very High
Queue records and logs are clearly displayed.	4.43	Very High

Table 2.8 Accessibility and Monitoring Evaluation

Table 2.8 presents the evaluation results for accessibility and monitoring features of the TagSakay administrative interface. The Very High ratings indicate that administrators were able to access queue information and monitor driver rotations with ease. Clear presentation of queue records and logs contributed to efficient oversight, enabling barangay personnel to track queue status accurately and make informed operational decisions.

Queue Processing

Indicator	Mean	Description
RFID scan records are immediately reflected on the dashboard.	4.49	Very High

Queue history and logs are accurate and complete.	4.45	Very High
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Table 2.9 Queue Processing Evaluation

Table 2.9 summarizes the effectiveness of queue processing on the administrator side. Results show that RFID scan records are reflected on the dashboard almost immediately, ensuring real-time visibility of queue movements. The accuracy and completeness of queue history logs further demonstrate the system's reliability in maintaining consistent records, which is essential for auditability and dispute resolution.

Report Generation

Indicator	Mean	Description
It is easy to generate official queue reports.	4.58	Excellent
Generated reports are clear and well-organized.	4.44	Very High

Table 2.10 Report Generation Evaluation

Table 2.10 highlights the system's capability to generate official queue reports. The Excellent and Very High ratings indicate that administrators

found report generation straightforward and the resulting documents clear and well-organized. These findings confirm that TagSakay supports effective documentation and reporting requirements at the barangay level.

System Control and Security

Indicator	Mean	Description
Role-based access control functions correctly.	4.51	Excellent
System data is protected from unauthorized access.	4.48	Very High

Table 2.11 System Control and Security Evaluation

Table 2.11 reflects the system's effectiveness in terms of access control and data security. Results indicate that role-based access control functions as intended, limiting system privileges based on user roles. Additionally, the Very High rating for data protection suggests that administrators have confidence in the system's ability to safeguard sensitive queue information from unauthorized access.

The evaluation results confirm that TagSakay successfully meets its design objectives. High ratings across all functional and non-functional requirements indicate that the system

improves queue fairness, operational transparency, and administrative oversight. The use of RFID technology significantly reduced manual intervention while ensuring accurate and real-time queue updates. These findings validate TagSakay's suitability for deployment in barangay-level tricycle queue management.

5.1 System Functionality and Performance Evaluation

5.1.1 Unit Testing Results

Unit testing was conducted to assess the core functionalities of the TagSakay RFID tapping system prior to system integration. These tests focused on the reliable logging of driver queue entries using RFID technology, the accuracy of real-time visual updates, and the responsiveness of the web-based administrative dashboard. The system was evaluated from the perspectives of two primary stakeholder groups: the tricycle drivers and the barangay transport staff.

From the drivers' side, unit tests confirmed the successful performance of the physical tap-in station. Testing validated that the ESP32-powered microcontroller and RFID reader could consistently capture driver information from the issued RFID cards. Real-time visual updates were successfully transmitted to the terminal's display screen, ensuring that drivers could immediately verify their position in the queue. While the system demonstrated high reliability in capturing timestamps, performance was noted to be dependent on a stable power source at the canopy-type terminal.

From the administrative side, unit testing focused on the web-based dashboard developed using Vue.js and CloudFlare Workers. Tests demonstrated that barangay staff could accurately view the live queue data and monitor driver rotations in real time. The PostgreSQL database successfully stored and retrieved structured records, allowing for the generation of daily and weekly reports. Overall, the unit tests verified that the automated logging of queue entries effectively replaced manual processes such as chalkboard listings and verbal coordination.

5.1.2 Integration Testing Results

Integration testing for the TagSakay system was conducted to ensure that the hardware components (RFID reader, ESP32, display) and the software modules (Web Dashboard, PostgreSQL database) interacted correctly. All integration tests were executed to verify that the flow from a physical "tap" to a digital record was seamless.

The registration process was tested by simulating the issuance of RFID cards to 40 registered tricycle drivers. The application correctly linked each unique RFID tag to a specific driver's profile in the database. When a card was tapped at the station, the system successfully retrieved the driver's info and updated the queue order across both the physical terminal display and the remote admin dashboard.

Testing also covered the synchronization between the physical tap station and the cloud-ready backend. It was confirmed that while the system functioned optimally with Wi-Fi or mobile data, the absence of

connectivity prevented real-time synchronization until a reconnection occurred. The report generation feature was also integrated, allowing administrators to filter and view rotation data, which effectively addressed the previous lack of centralized digital records.

Feature	Test Description	Result
RFID Tap-In	Verified that physical RFID taps are captured and logged with accurate timestamps.	Passed
Driver Registration	Confirmed that RFID cards are correctly linked to driver profiles in the database.	Passed
Real-Time Display	Validated that the terminal display updates instantly after a successful tap.	Passed
Admin Dashboard	Ensured the web interface reflects the current queue status from the backend.	Passed
Report Generation	Tested the automated creation of daily and weekly rotation monitoring reports.	Passed

Connectivity Sync	Verified data synchronization between the ESP32 station and the PostgreSQL database.	Passed
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Table 2.12 Integration Testing Summary for TagSakay System

5.2 Performance Testing and System Reliability

5.2.1 System Responsiveness

System responsiveness was evaluated by observing the latency between a physical RFID tap and the appearance of the driver's name on the terminal display and admin dashboard. During standard testing, the system was generally responsive, completing the cycle in near real-time. However, minor delays were observed when the system relied on mobile data in areas with variable signal strength. These variations were attributed to network traffic rather than hardware inefficiencies.

5.2.2 Stress and Load Testing

Stress testing was conducted by simulating high-traffic periods, such as peak hours, where multiple drivers might arrive at the terminal simultaneously. While the system remained stable, it was noted that the RFID reader requires sequential tapping; if two cards are presented at the exact same moment, one may not be read. The system remained functional without crashes, but the researcher emphasized that drivers must consistently comply with the

one-at-a-time tap-in procedure to ensure data integrity.

5.3 Usability Testing and User Feedback

5.3.1 User Acceptance Testing (UAT) Results

UAT was conducted with tricycle drivers from CATICOMODA Route A-29-A and barangay personnel. Drivers found the physical tap process much simpler than verbal coordination, noting it reduced ambiguity and disputes regarding their turn. Barangay staff appreciated the visual display and the ability to audit queue entries without relying on memory or chalkboards.

5.3.2 Qualitative Feedback

Testers praised the low-cost and robust nature of the RFID cards compared to smartphone-based solutions, which would require data plans and expensive hardware. However, feedback suggested the need for a more rugged housing for the tap station to withstand outdoor conditions at the terminal. Drivers also requested a "tap-out" feature for future iterations to track when they have left the queue.

5.4 Security Assessments

Security measures included the use of unique RFID card IDs that are hard-coded to specific drivers, preventing unauthorized users from entering the queue. The web dashboard utilized secure login credentials for barangay staff to ensure that only authorized personnel could edit driver roles or delete records. The physical "tap" itself provides a verifiable digital trail that

discourages fraudulent queue entries and enhances administrative accountability.

5.5 Limitations and Areas for Improvement

A primary limitation of the TagSakay system is its dependency on stable internet and power for real-time synchronization. Without connectivity, the admin dashboard cannot update until the station reconnects. Additionally, the current version lacks automated "tap-out" logging and does not include advanced features like GPS tracking or facial recognition. Future iterations should focus on integrating a local storage cache for offline taps and exploring vision-based systems for automatic vehicle identification.

5.6 Comparison with Related Systems

Several technological solutions have been explored in the Philippines to address the complexities of public transportation, particularly for tricycle and jeepney operations. These systems generally focus on fare-based management or remote tracking to modernize the local transport sector. Notable examples include the NodeMCU-Based Contactless Fare Management System for tricycle commuters and the Jeepney Queuing Tracking System (JEEPQTS). While these systems demonstrate the feasibility of using low-cost hardware like Arduino and NodeMCU, they primarily target financial transactions or matching drivers with passenger demand via SMS, rather than enforcing physical queue order.

Unlike these related systems, which are often built around revenue collection and ticketing, TagSakay is designed specifically for queue rotation and monitoring. For instance, while the Internet of Things-Based Revenue Collection System and various Automatic Fare Collection (AFC) systems focus on reducing revenue leakage and calculating distance-based fares, they often lack an on-ground visual enforcement mechanism for driver turn-taking. Furthermore, existing queue management tools, such as the Queuing Management System for Lingayen Jeepney Terminal, focus on administrative record-keeping but lack the physical queue enforcement provided by TagSakay's hardware-software integration.

TagSakay introduces a key enhancement by prioritizing a low-cost, physical tap-in solution using RFID technology. While many international and local systems rely heavily on GPS for tracking, these often encounter signal reliability issues in shaded terminal areas and require expensive data plans for drivers. By utilizing passive RFID tags and a dedicated local display, TagSakay ensures that queue positions are transparently communicated to all drivers in real-time without the overhead of fare-tracking or complex smartphone requirements. This approach effectively fills a critical gap in on-ground queue enforcement for grassroots-level communities where infrastructure-independent and cost-effective solutions are essential.

CHAPTER 6 - RECOMMENDATION AND CONCLUSION

6.1 Recommendation

Based on the findings and experience gained during the development of TagSakay, several improvements are proposed to enhance the system's security, scalability, and functional depth. These recommendations aim to reinforce the efficiency of tricycle terminal operations and serve as a foundation for future research in automated transport management.

Firstly, as digital infrastructures in the Philippines face increasing risks of unauthorized access, the researcher highly recommends strengthening the security of the web dashboard and database. Enhanced enforcement of the Confidentiality, Integrity, and Availability (CIA) triad—specifically through encrypted data transmission between the ESP32 microcontroller and the PostgreSQL database—is essential to protect driver data from potential tampering.

Secondly, while the current system utilizes physical RFID tags, future iterations should explore the integration of machine learning or computer vision. Implementing License Plate Recognition (LPR) could serve as a secondary validation layer to ensure the vehicle entering the queue corresponds with the registered RFID card, thereby preventing card-sharing or fraudulent entries. Additionally, the inclusion of SMS notifications could alert drivers when they are nearing the top of the queue, allowing them to wait in shaded areas away from the terminal without the risk of missing their turn.

The existing prototype relies heavily on stable Wi-Fi or mobile data for real-time synchronization. Future

research should investigate alternative long-range communication technologies, such as LoRa (Long Range), to enable the tap stations to communicate with the central server over several kilometers independent of conventional internet infrastructure. This would be particularly beneficial for terminals located in network "dead zones."

Furthermore, the system's scope is currently limited to a specific terminal route in Davao City. There is significant potential for expanding TagSakay's coverage to other tricycle, jeepney, or taxi terminals. To support this, the implementation of geofencing functions is suggested to ensure that "taps" are only valid when the device is physically located within designated terminal boundaries. Future versions could also feature a "Passenger App" to allow commuters to view queue lengths and estimate waiting times.

To address hardware durability in outdoor environments, the researcher proposes the development of weather-resistant, industrial-grade housing for the RFID stations. Moreover, establishing clear policies and automated tracking for "tap-outs" (instances where a driver leaves the queue for a break) would further refine the accuracy of the live rotation data.

Finally, the study recommends the strategic integration of this system with local government transport offices. By providing real-time data on driver activity and terminal congestion, TagSakay can serve as a foundational tool for data-driven urban transport planning and municipal policy-making.

6.2 Conclusion

TagSakay is an automated queue monitoring system developed to facilitate orderly rotations and transparent coordination between tricycle drivers and barangay transport authorities. By leveraging a web-based platform and an RFID-integrated mobile station, the system successfully replaces manual, error-prone methods such as chalkboards and verbal agreements with a precise digital record.

A primary objective of the system was to provide drivers and dispatchers with a real-time visual representation of the queue. This was achieved through a synchronized display at the terminal and a remote administrative dashboard. This functionality minimizes disputes among drivers and ensures a "first-in, first-out" policy is strictly and fairly maintained.

Another key objective was the automation of driver rotation logs for administrative reporting. The system successfully facilitated the digital capture of every "tap," allowing barangay personnel to generate daily and weekly reports efficiently. This centralized data management enables more effective monitoring of driver compliance and provides a verifiable audit trail for terminal operations.

The system also addressed the need for low-cost, accessible technology within the local transport sector. By utilizing affordable components like the

ESP32 and passive RFID tags, TagSakay offers a sustainable alternative to expensive tracking systems. The decentralized nature of the portable tap station ensures it can be deployed quickly at various terminal types with minimal infrastructure requirements.

The results of this research demonstrate the practical utility of integrating hardware and software for transport management. The system garnered strong approval during testing, proving its reliability, ease of use, and effectiveness in reducing terminal congestion. The automation of the queue process not only saves time for drivers but also professionalizes the operations of the local transport unit.

In conclusion, the research verifies that a low-cost RFID-based system like TagSakay can significantly enhance the efficiency of tricycle queue rotations at the local level. While there are opportunities for further refinement, such as adding offline local storage and expanding to cross-platform mobile environments, the overall system is effective in attaining its desired goals. Through ongoing adoption and refinement, TagSakay can contribute to making community transport systems more organized, transparent, and technologically prepared for future modernization.

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APPENDICES

APPENDIX A. FRAMEWORK, FLOWCHART, AND DIAGRAMS

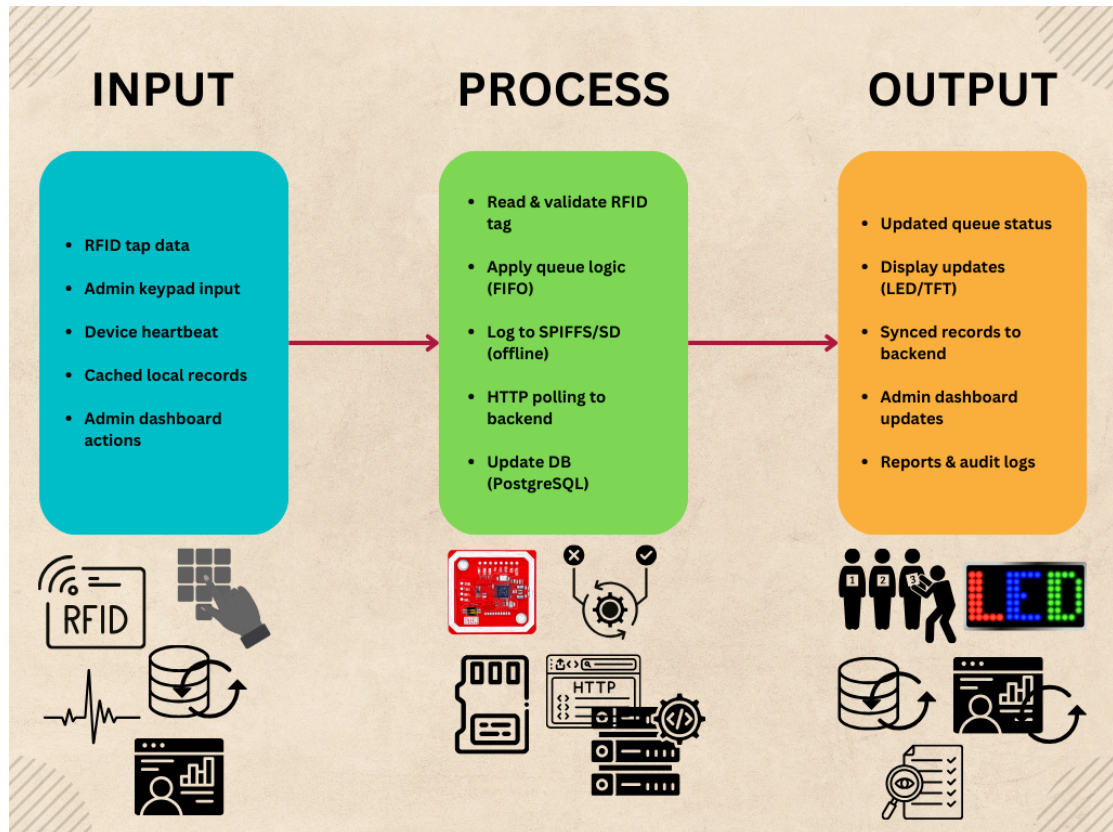


Figure 1.0 Conceptual Framework

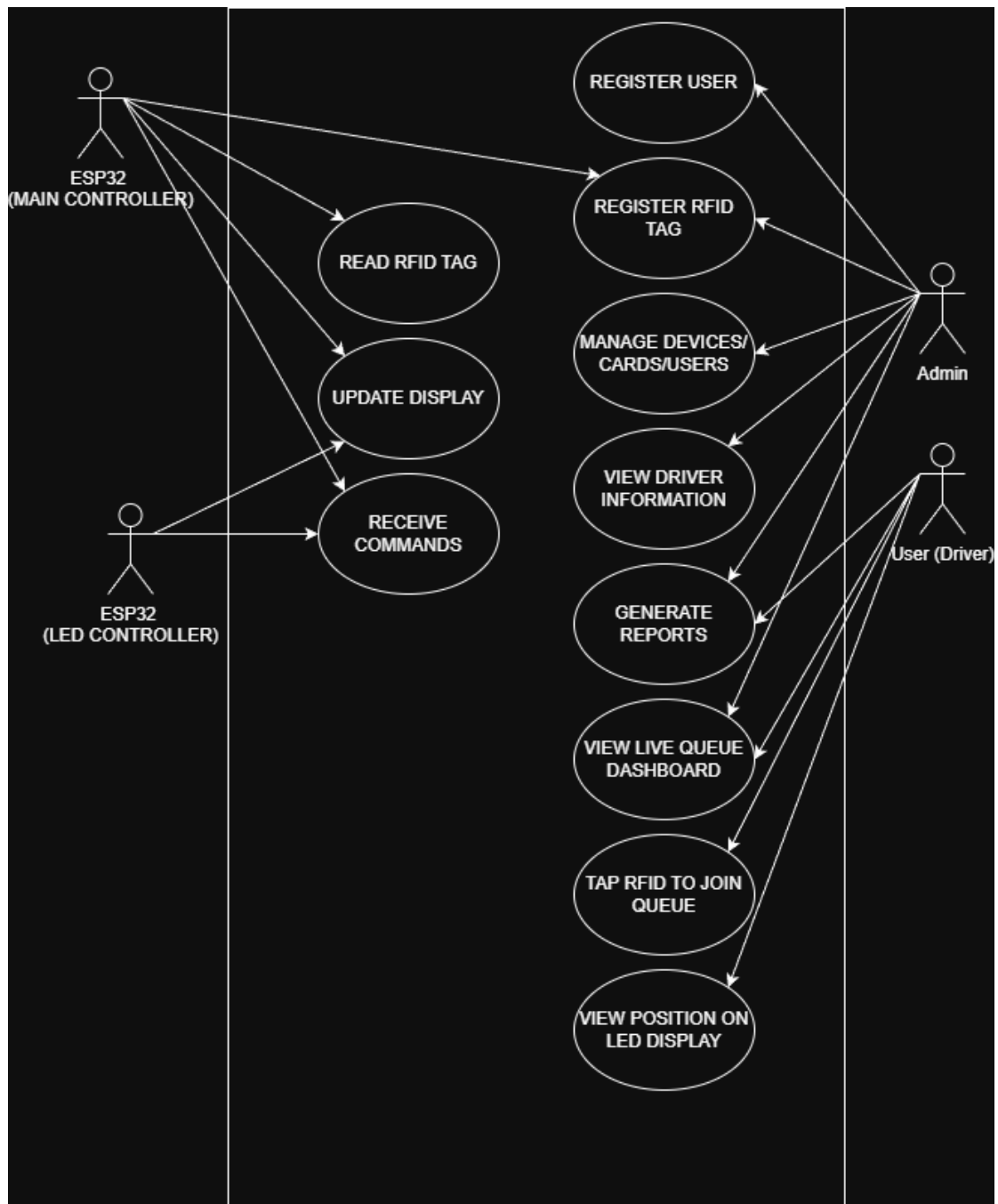


Figure 1.1 Use Case Diagram

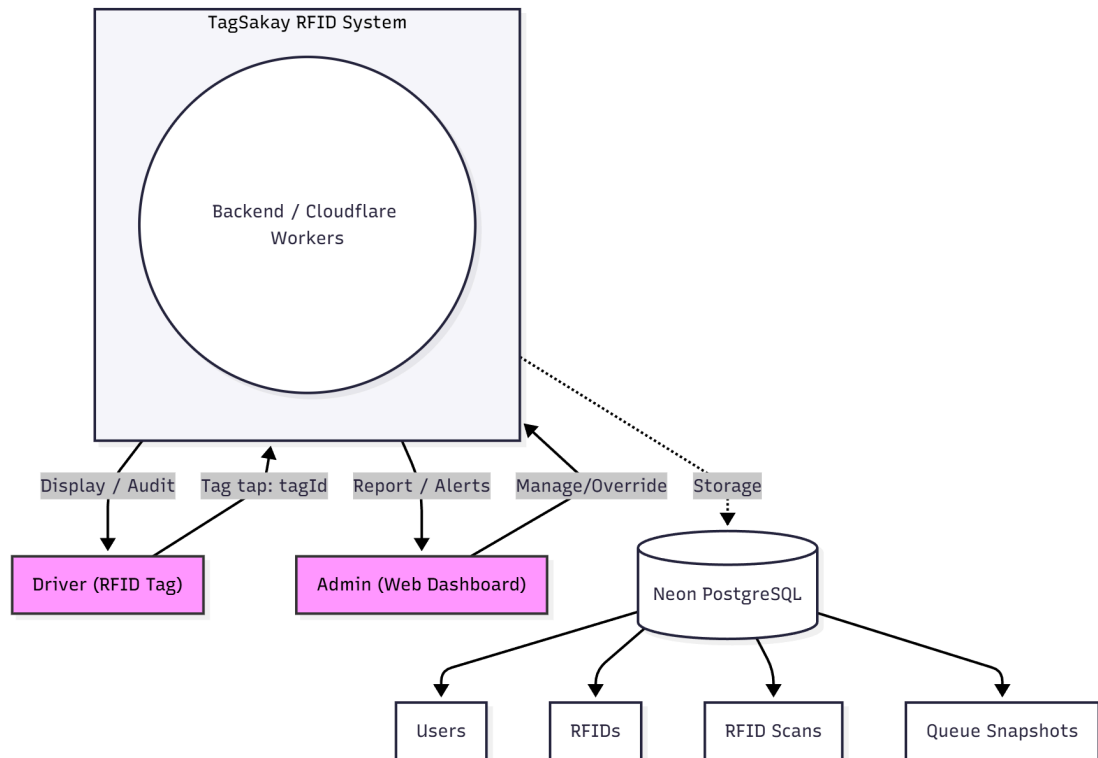
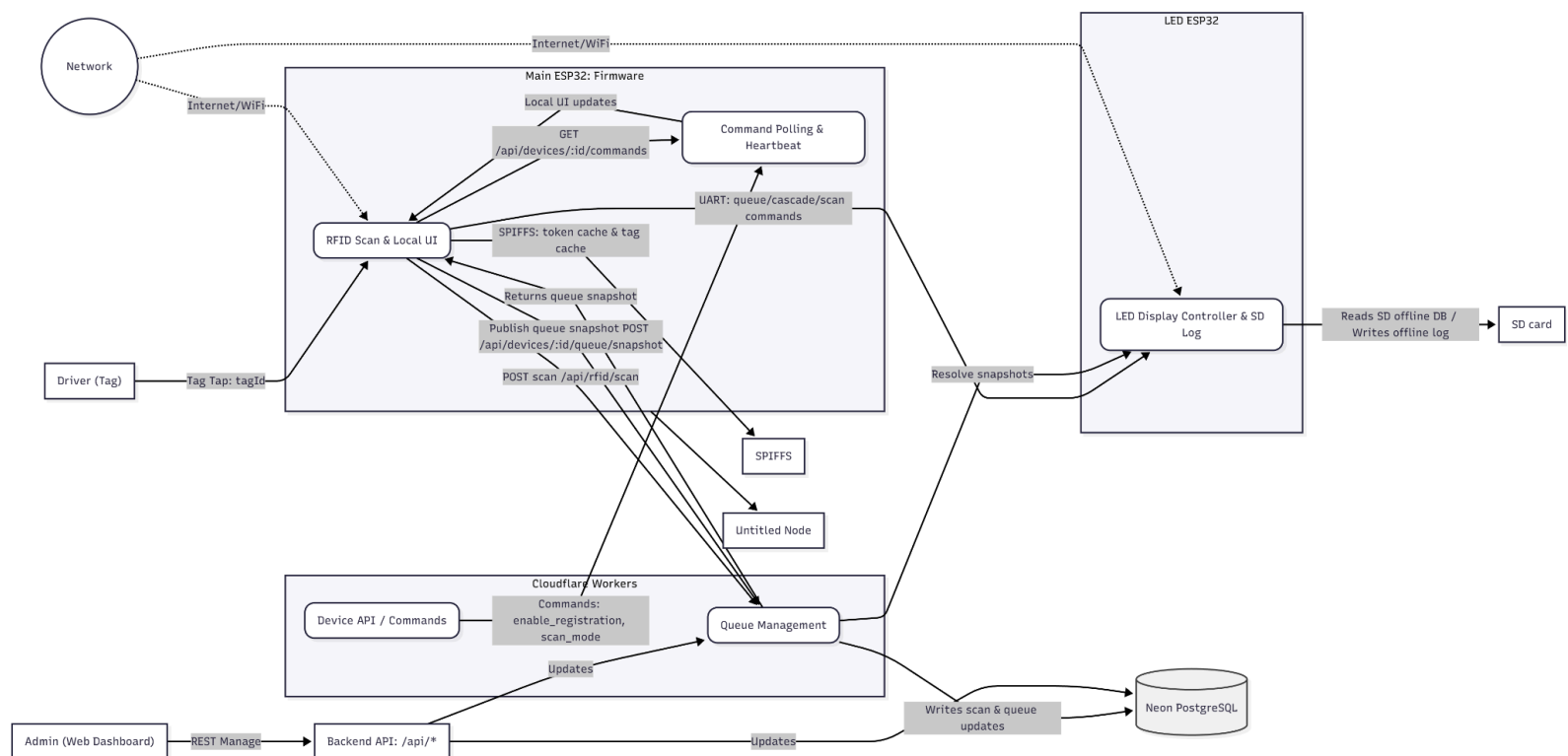


Figure 1.2 and Figure 1.3 - Data Flow Diagram 0 and 1



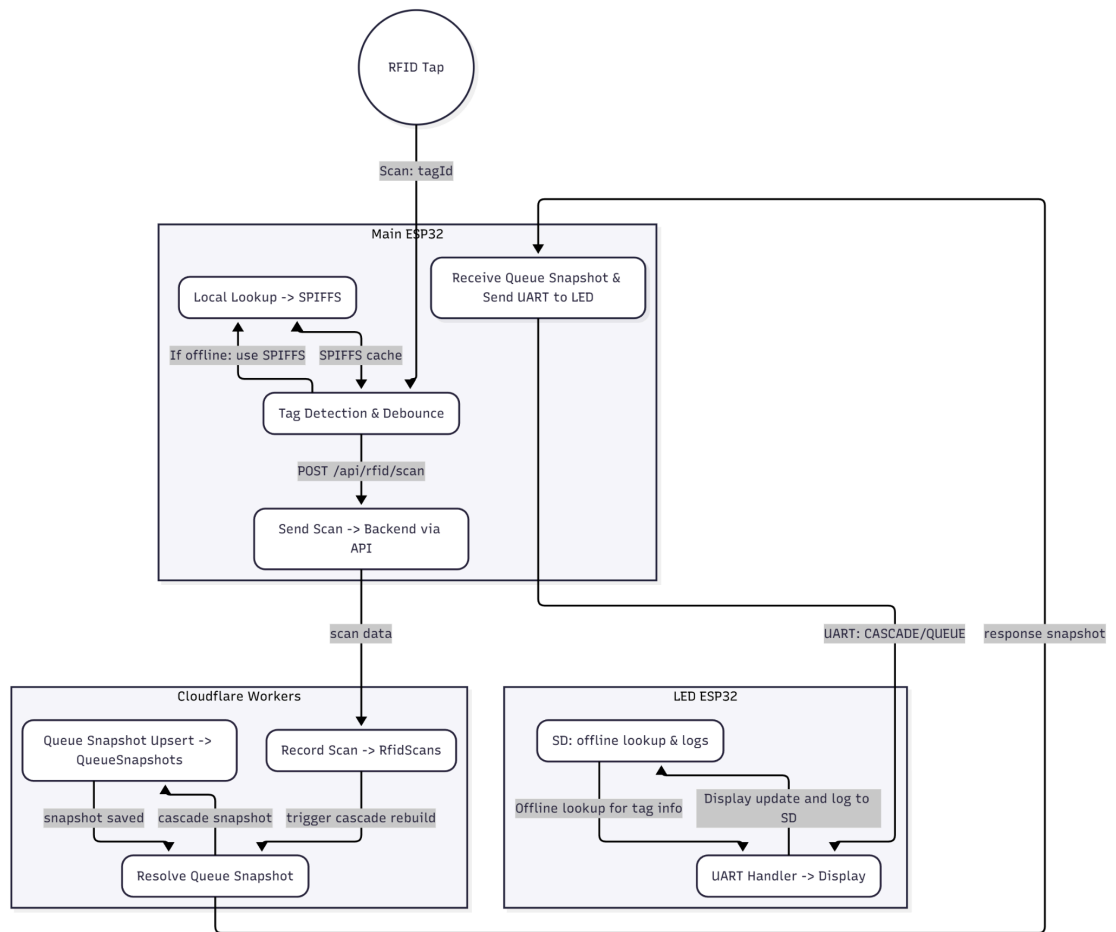


Figure 1.4 Data Flow Diagram 2

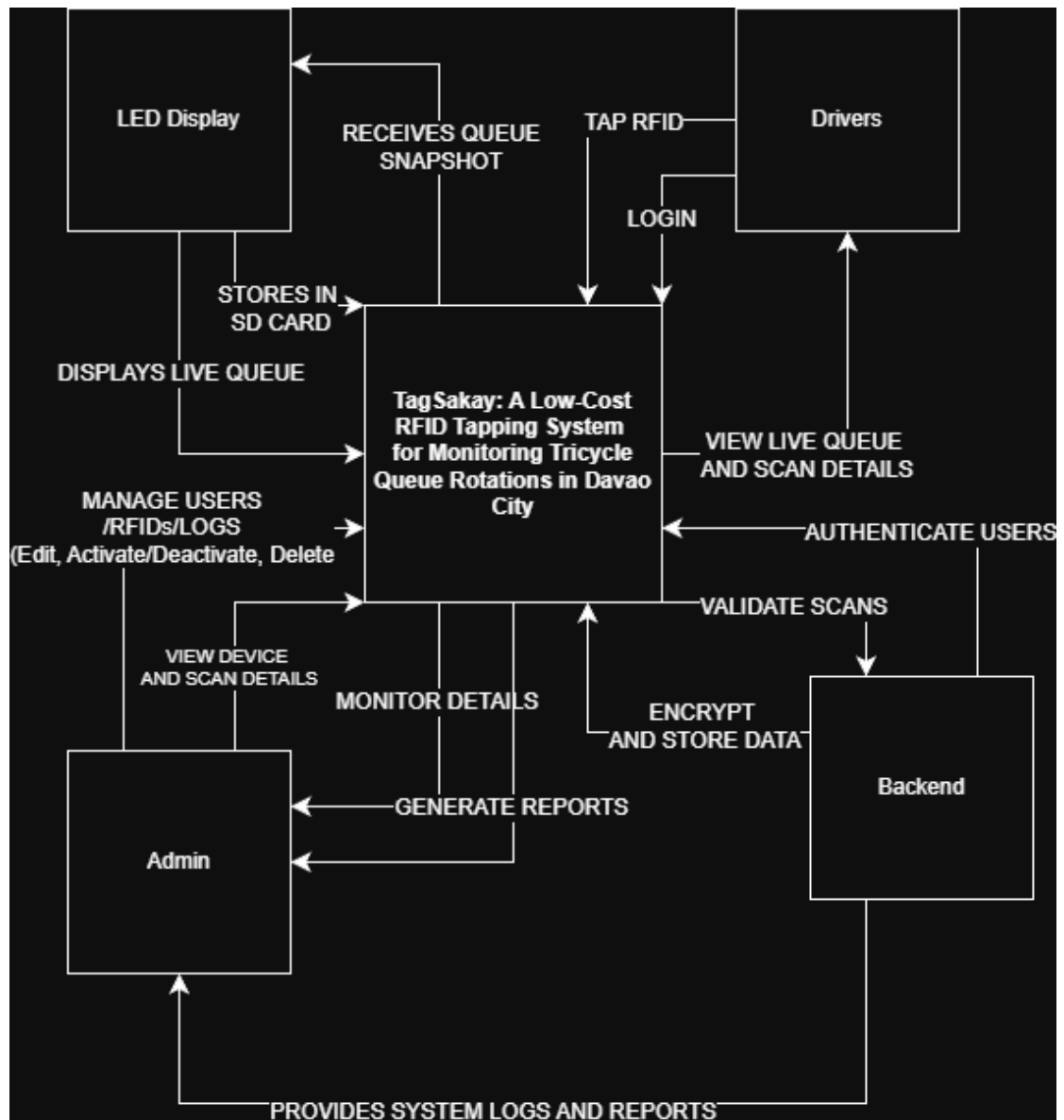


Figure 1.5 Context Flow Diagram

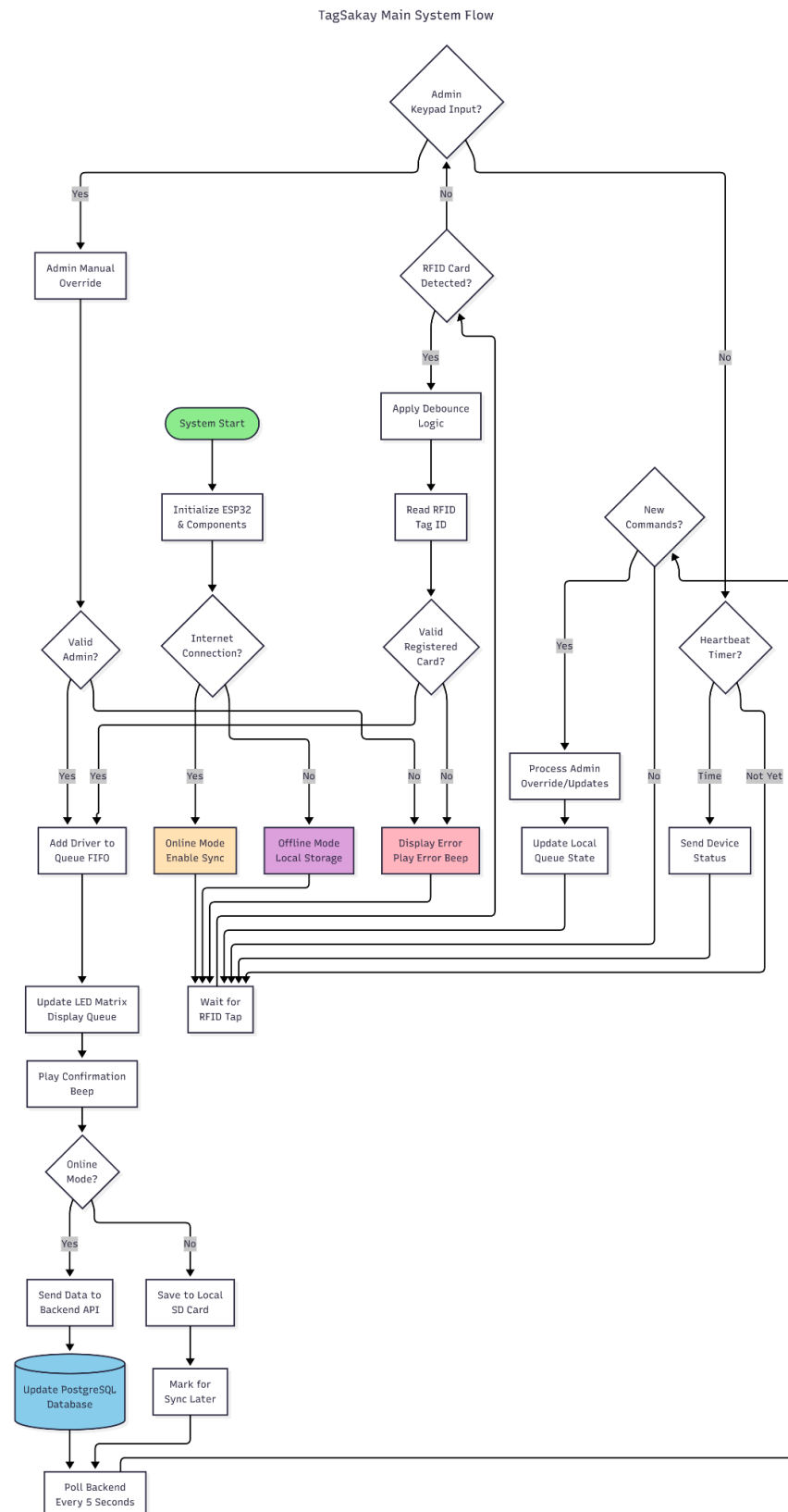


Figure 1.6 System Flowchart (Firmware)

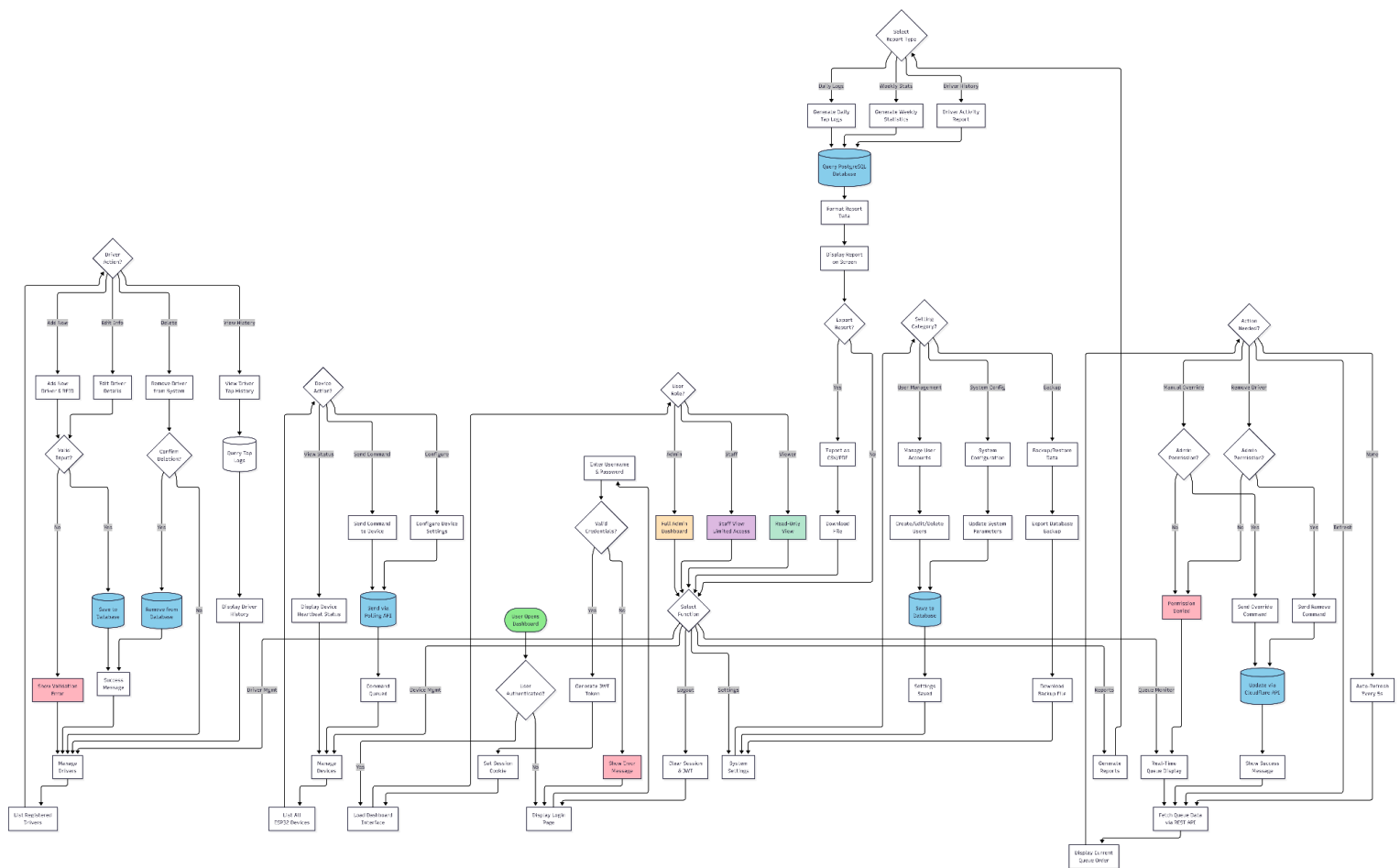


Figure 1.7 System Flowchart (Webview)

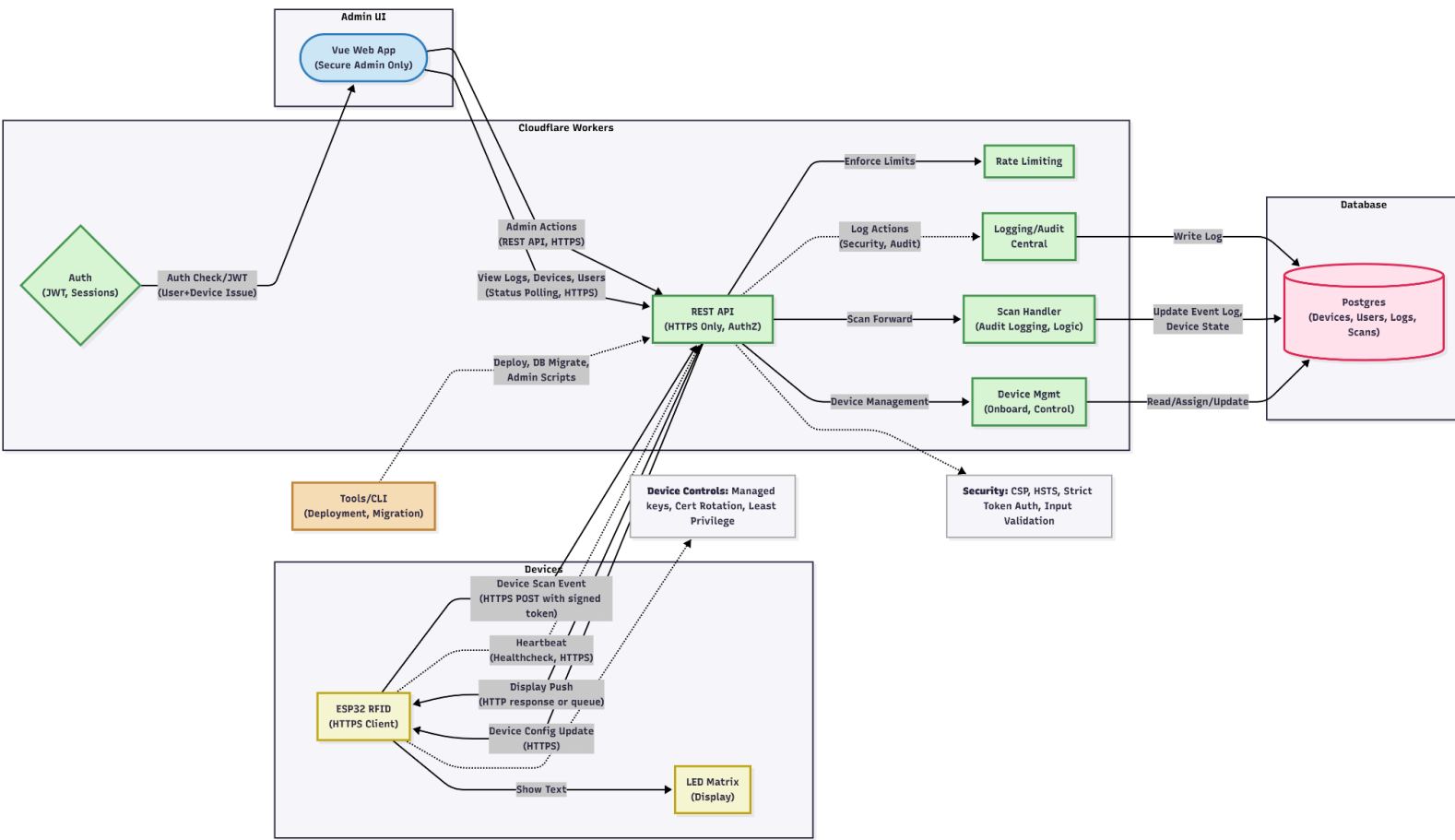


Figure 1.8 System Architecture